Research Note 5

1 code

```
ClearAll["Global`*"]


limit = 4578.4;       (*GeV^-3*)
v = 10^-3;
mchi = 0.01;          (*10 MeV=0.01 GeV*)
alphaXDerived[R_] :=
  Sqrt[(limit*mchi^3*v^4)/(4*Pi*
      R^2*(R^2/(R^2 + 1) + Log[1/(R^2 + 1)]/(R^2 + 2)))];

alphaXB3[R_] := 0.02*Sqrt[10^-6/mchi]*((mchi*v)/(R*10^-3))^2;

alphaXB4[R_] := 10^-10*(mchi/10^-3)^(3/2);

alphaXB2[R_] :=
  Sqrt[(limit*mchi^3*v^4)/(8*Pi*(Log[1 + R^2] - R^2/(1 + R^2)))];

(*Slope Verification)*)

(*slope=d(lnalpha)/d(lnR)=(1/alpha dalpha)/(1/R dR)=R/alpha*dalpha/dR*)

derivedSlope[R_] =
  Simplify[(R*D[alphaXDerived[R], R])/alphaXDerived[R]];
b3Slope[R_] = Simplify[(R*D[alphaXB3[R], R])/alphaXB3[R]];
b4Slope[R_] = Simplify[(R*D[alphaXB4[R], R])/alphaXB4[R]];
b2Slope[R_] = Simplify[(R*D[alphaXB2[R], R])/alphaXB2[R]];

Print["R -> 0   my   result slpoe: ", Limit[derivedSlope[R], R -> 0]];
Print["R -> 0 B3 slope: ", Limit[b3Slope[R], R -> 0]];
Print["R -> 0 B2 slope: ", Limit[b2Slope[R], R -> 0]];
Print["R -> Infinity, B2 slope: ", Limit[b2Slope[R], R -> Infinity]];
Print["\n R -> Infinity, my result slope: ",
  Limit[derivedSlope[R], R -> Infinity]];
Print["R -> Infinity, B4 slope: ", Limit[b4Slope[R], R -> Infinity]];




LogLogPlot[{alphaXDerived[R], alphaXB3[R], alphaXB4[R]}, {R, 10^-3,
  10^3}, PlotStyle -> {Directive[Thick, Blue],
   Directive[Thick, Dashed, DarkGreen],
   Directive[Thick, DotDashed, Red]      }, Frame -> True,
 FrameLabel -> {Style[
    "Dimensionless Parameter R (v * \!\(\*SubscriptBox[\(m\), \
\(\[Chi]\)]\) / \!\(\*SubscriptBox[\(m\), \(\[Phi]\)]\))", 12, Bold],
```

```
   Style["Coupling Constant \!\(\*SubscriptBox[\(\[Alpha]\), \(\[Chi]\)
      ]\)",
     12, Bold]}, GridLines -> Automatic,
 Filling -> {1 -> Top, 2 -> Top, 3 -> Top},
 FillingStyle -> Directive[Gray, Opacity[0.05]],
 PlotLegends ->
  Placed[LineLegend[{"My Derivation (\!\(\*SubscriptBox[\(\[Sigma]\), \
\(T2\)]\))", "arXiv B3 Bound (R << 1)", "arXiv B4 Bound (R >> 1)"},
    LabelStyle -> {FontSize -> 11}], {0.65, 0.75} ],
 PlotLabel ->
  Style["SIDM Bounds & Asymptotic Behavior Verification", 14, Bold],
 ImageSize -> 550]



alphaMax[R_] :=
  Max[alphaXDerived[R], alphaXB2[R], alphaXB3[R], alphaXB4[R]];

LogLogPlot[{alphaXDerived[R], alphaXB2[R], alphaXB3[R], alphaXB4[R],
  alphaMax[R]}, {R, 10^-3, 10^3},
 PlotStyle -> {Directive[Thick, Blue],
   Directive[Thick, Orange, Dashed],
   Directive[Thick, Dashed, DarkGreen],
   Directive[Thick, DotDashed, Red], None}, Frame -> True,
 FrameLabel -> {Style[
    "Dimensionless Parameter R (v * \!\(\*SubscriptBox[\(m\), \
\(\[Chi]\)]\) / \!\(\*SubscriptBox[\(m\), \(\[Phi]\)]\))", 12, Bold],
   Style["Coupling Constant \!\(\*SubscriptBox[\(\[Alpha]\), \(\[Chi]\)
      ]\)",
     12, Bold]}, GridLines -> Automatic,
 Filling -> {5 -> Top},
 FillingStyle -> Directive[Gray, Opacity[0.5]],
 PlotLegends ->
  Placed[LineLegend[{"My Derivation (\!\(\*SubscriptBox[\(\[Sigma]\), \
\(T2\)]\))", "1709.07882 Eq (B2)", "arXiv B3 Bound (R << 1)",
     "arXiv B4 Bound (R >> 1)"},
    LabelStyle -> {FontSize -> 11}], {0.7, 0.25}],
 PlotLabel ->
  Style["SIDM Bounds & Asymptotic Behavior Verification w/ B2", 14,
   Bold], ImageSize -> 600]
alphaRatio[R_] := alphaXB2[R]/alphaXDerived[R];
Print["R -> Infinity, Ratio (B2/Derived): ",
  Limit[alphaRatio[R], R -> Infinity]];
```

2 results

In this code, I add the B2 into the previous plot. B2's behavior matches with B3 & B4.

I made the shadow fills the highest area. Since my slope is -1 at R>>1, the ratio is predictable to be infinity.
$R \to 0$, my result slope:-2.
$R \to 0, B3$ slope: -2.
$R \to 0, B2$ slope: -2.
$R \to \infty, B2$ slope: 0.
$R \to \infty$, my result slope: -1.

$R \to \infty, B4$ slope: 0.

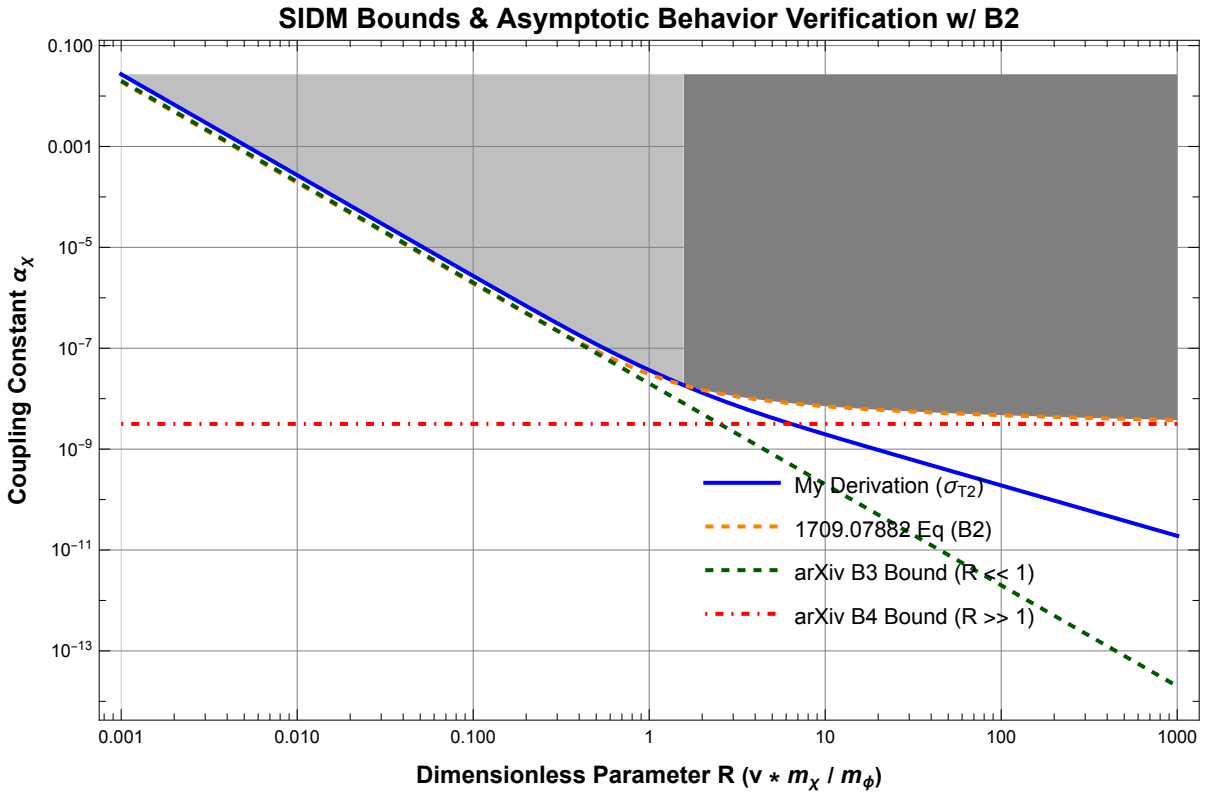


Figure 1:

$R \to \infty$, Ratio $\frac{B2}{Derived} : \infty$

My result goes to 10^{-11}, while the paper's result goes to 10^{-9}, since I used mchi= 0.01 GeV =10 MeV. Instead of ? 100MeV.

Since the ratio is infinity, I found the ratio's full form to make it clear.

```
C0 = limit*mchi^3*v^4

alphaXDerived[R_] :=
  Sqrt[C0/(4*Pi*R^2*(R^2/(R^2 + 1) + Log[1/(R^2 + 1)]/(R^2 + 2)))];
alphaXB2[R_] := Sqrt[C0/(8*Pi*(Log[1 + R^2] - R^2/(1 + R^2)))];

ratioExpr = alphaXB2[R]/alphaXDerived[R];


simplifiedRatio = FullSimplify[ratioExpr, R > 0];

Print["The simplified ratio alpha_B2 / alpha_Derived is:"];
Print[simplifiedRatio];

Print["\nAsymptotic behavior for R -> Infinity:"];
Print[Series[simplifiedRatio, {R, Infinity, 1}] // Normal]
```

and got results:

The simplified ratio α_{B2} / $\alpha_{Derived}$ is:

$$0.707107 * R * \sqrt{\frac{\frac{R^2}{R^2+1} - \frac{\log(R^2+1)}{R^2+2}}{\frac{1}{R^2+1} + \log(R^2+1) - 1}}$$

The Asymptotic behavior for $R \to \infty$ is (to R^{-1}):

$$0.707107 * R * \sqrt{\frac{1}{2\log(R)-1}} - \frac{1.41421\left(\frac{1}{2\log(R)-1}\right)^{3/2}\left(1.\log^2(R)+0.25\right)}{R}$$

2.1 Comments

As I mentioned in Note 3, for light mediator limit, there is a logorism behavior but very small. However, I kept it for any potential important logorism behavior.

Another thing is I didn't use viscosity cross section.

2.2 Viscosity Cross Section

I used my note 4.5's result for the mother formula and do the same thing:

```
    ClearAll["Global`*"];


limit = 4578.4; (*GeV^-3*)
v = 10^-3;
mchi = 0.01; (*10 MeV*)

fViscosity[R_] := (R^4 + 2 R^2 + 2)/(R^2 (R^2 + 2))*Log[R^2 + 1] - 1;
alphaXViscosityMother[R_] := Sqrt[(limit*mchi^3*v^4)/(Pi*fViscosity[R])
   ];

fB2[R_] := Log[1 + R^2] - R^2/(1 + R^2);
alphaXB2[R_] := Sqrt[(limit*mchi^3*v^4)/(8*Pi*fB2[R])];

alphaMax[R_] := Max[alphaXViscosityMother[R], alphaXB2[R]];

LogLogPlot[{alphaXViscosityMother[R], alphaXB2[R], alphaMax[R]}, {R,
  10^-3, 10^3},
 PlotStyle -> {Directive[Thick, Blue],
   Directive[Thick, Orange, Dashed], None}, Frame -> True,
 FrameLabel -> {Style[
    "Dimensionless Parameter R (v * \!\(\*SubscriptBox[\(m\), \
\(\[Chi]\)]\) / \!\(\*SubscriptBox[\(m\), \(\[Phi]\)]\))", 12, Bold],
   Style["Coupling Constant \!\(\*SubscriptBox[\(\[Alpha]\), \(\[Chi]\)
      ]\)",
     12, Bold]}, GridLines -> Automatic, Filling -> {3 -> Top},
 FillingStyle -> Directive[Gray, Opacity[0.3]],
 PlotLegends ->
  Placed[LineLegend[{"Note 4.5 Mother Formula (\!\(\*SubscriptBox\
[\(\[Sigma]\), \(V2\)]\))", "arXiv 1709.07882 Eq (B2)"},
    LabelStyle -> {FontSize -> 11}], {0.65, 0.25}],
 PlotLabel ->
  Style["Comparison: Viscosity Mother Formula vs. B2", 14, Bold],
 ImageSize -> 600]


ratioViscosity[R_] := alphaXB2[R]/alphaXViscosityMother[R];
Print["R -> Infinity, Ratio (B2 / Viscosity Mother): ",
  Limit[ratioViscosity[R], R -> Infinity]];
```

2.3 results

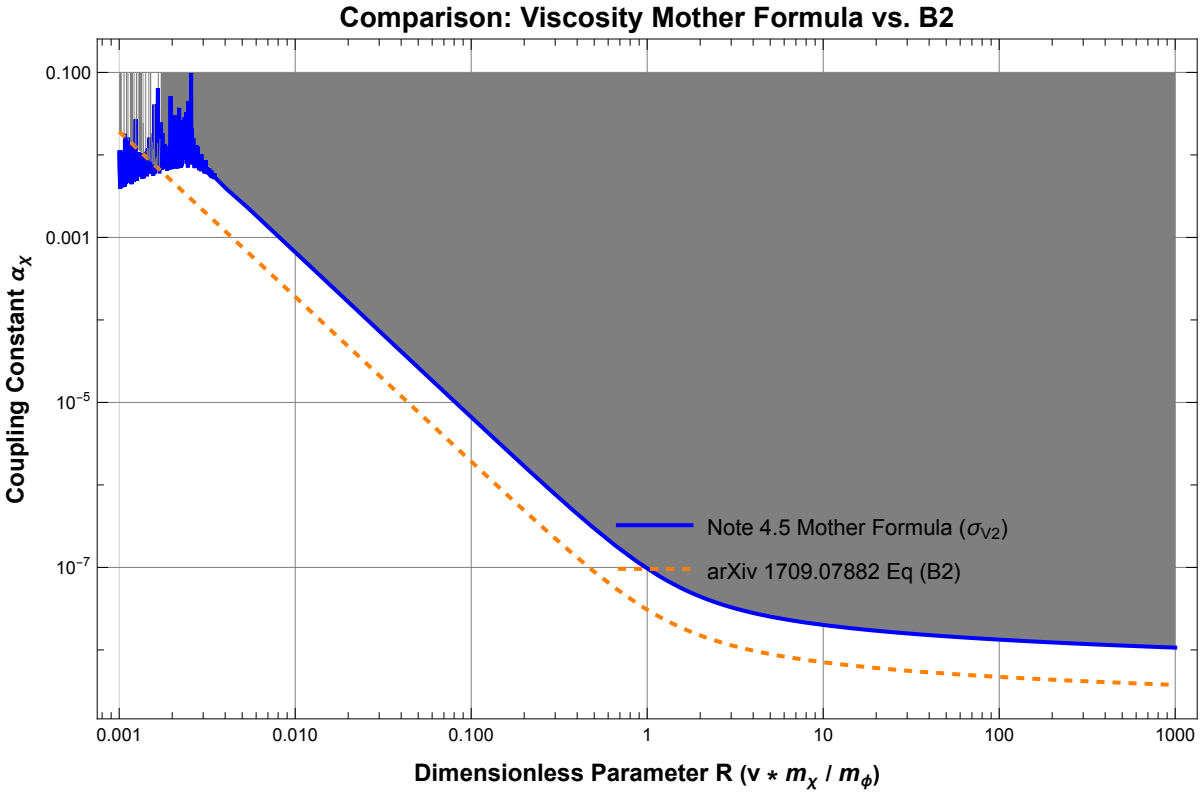


Figure 2:

$R \to \infty$, Ratio (B2 / Viscosity Mother): $0.353553 = \frac{1}{\sqrt{8}}$

In the paper, B2, the coefficient is 8π, my mother formula's coefficient is π.

$$\sigma_{TB2} \propto 8\pi\alpha_\chi^2, \tag{1}$$
$$\sigma_V \propto \pi\alpha_\chi^2 \tag{2}$$
$$\alpha_\chi \propto g_\chi^2 \tag{3}$$
$$limit \propto \sigma \tag{4}$$
$$\tag{5}$$

so

$$8\pi\alpha_\chi^2 \propto limitTB2 \tag{6}$$
$$\pi\alpha_\chi^2 \propto limitV \tag{7}$$

so

$$\alpha_\chi \propto \sqrt{\frac{1}{8\pi}}(Transfer) \tag{8}$$
$$\alpha_\chi \propto \sqrt{\frac{1}{\pi}}(Viscosity) \tag{9}$$

With a ratio $\frac{1}{\sqrt{8}}$

In note 2 I think for process 1, the paper and I are the same, but for process 2, 3, because of identical particles, i argued there should be an 1/2 factor. Which is shown in the plot Figure 1, at $R \ll 1$ limit, there is an $\frac{1}{\sqrt{2}}$ difference. But same slope. However for $R \gg 1$, the slope is different. So I tried to use viscosity cross section. Now there is just an $\frac{1}{\sqrt{8}}$ difference. Note that the paper is using transfer cross section and did some unknown methods, while I am using viscosity cross section. This might be the key.

Another thing I worry about is some vs could be expressed in R, or some vs inside R could be moved out and makes the final limit expression different. Now I followed substitute $M_A \to \frac{m_\chi v}{R}$ and then expand at $v = 0$ to 0th order and then expand at R=0 or ∞ at 4th order.

More generally, I don't know when to take the $v \to 0$ limit. Should it be before R limit or after?